

Unit-1:

1. Differentiate between supervised learning and unsupervised learning.
2. Explain different data types used in modern machine learning paradigm with examples.
3. How do you select the appropriate regression model for a given problem?
4. Explain Gradient Descent and its role in linear regression.
5. What are some common applications of predictive modeling in various industries?
6. Name three key algorithms used in predictive modeling.
7. How do you evaluate the accuracy of a predictive model?
8. What is the difference between continuous and categorical variables?
9. How can ordinal variables impact regression analysis?
10. Why is it important to distinguish between qualitative and quantitative variables in data analysis?
11. What is regression analysis, and why is it used in data science?
12. How is regression analysis used to predict outcomes?

Unit-2:

1. Why use Logistic Regression?
2. How to construct Regression tree and write procedure to construct regression tree with example.
3. Compare and contrast the approaches used in linear regression and logistic regression to model and predict outcomes. Specifically, address the following:
 - How do the modeling approaches differ when using one-variable versus multiple-variable linear regression, and how does gradient descent contribute to optimizing these models? Provide examples where these models are applicable.
 - Discuss the impact of polynomial regression on linear regression models, including the potential risks of overfitting and how regularization can be applied to mitigate these risks.
 - In the context of logistic regression, explain how the maximum likelihood estimation (MLE) algorithm is used to estimate model parameters and how it differs from the approach used in linear regression.
 - Explore the strategies for handling multi-class classification problems in logistic regression. How do these strategies address the challenges of classifying data into more than two categories?
 - Examine how overfitting manifests in both linear and logistic regression models. What are the key differences in handling overfitting between these two types of regression, and how do regularization techniques play a role in each?
6. How do you select the appropriate regression model for a given problem?
7. What is the purpose of the gradient descent algorithm in linear regression?
8. What is linear regression, and how is it different from other types of regression?
9. How is the equation for linear regression with one variable expressed, and what do the terms represent?
10. Explain the concept of a cost function in linear regression.
11. How does learning rate impact the performance of gradient descent?
12. What is the principle of the maximum likelihood estimation in logistic regression?

13. How is the likelihood function defined for logistic regression?
14. How does gradient descent optimize the likelihood function in logistic regression?
15. What is overfitting, and why is it a problem in regression models?
16. What are the potential risks of using a high-degree polynomial regression model?

Unit-3 :

1. What are the Metrics used to validate the result of regression and explain each.
2. Explain CART (Classification & Regression tree) algorithm with example.
3. What is a hyperparameter? How do you choose which settings are going to be hyperparameters and which are going to be learnt? (either difficult to optimize or not appropriate to learn – learning.
4. model capacity by learning the degree of a polynomial or coefficient of the weight decay term always results in choosing the largest capacity until it overfits on the training set).
5. You are working on a time series data set. Your manager has asked you to build a high accuracy model. You start with the decision tree algorithm, since you know it works fairly well on all kinds of data. Later, you tried a time series regression model and got higher accuracy than decision tree model. Can this happen? Why?
6. You created a predictive model of a quantitative outcome variable using multiple regressions. What are the steps you would follow to validate the model?
7. What is overfitting a regression model? What are ways to avoid it?
8. How to check if the regression model fits the data well?
9. What is the principle of ordinary least square in linear regression?
10. Which evaluation metrics cannot be applied in case of logistic regression output to compare with target?
11. Which algorithm is used for Variable Selection?
12. How can we visually assess the bias-variance tradeoff in a learning curve?
13. What is the relationship between model complexity and the bias-variance tradeoff?
14. What is cross-validation, and why is it important in regression modeling?
15. How does k-fold cross-validation help improve model performance?

Unit- 4

1. How is regression analysis used in marketing to predict customer behavior?
2. Explain how regression analysis can assist in financial forecasting.
3. What role does regression analysis play in environmental studies?
4. How is regression analysis applied in transportation and logistics for demand prediction?
5. How do ensemble methods like stacking and boosting enhance regression model performance?

Case Studies:

1. Predicting Employee Salary
2. Education System
3. Stock Market
4. Real estate Market